

Laws for Work-Hardening and Low-Temperature Creep¹

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The true stress-strain curves of polycrystalline aluminum, copper, and stainless steel are shown to be adequately represented by an exponential approach to a saturation stress over a significant range. This empirical law, which was first proposed by Voce, is expanded to describe the temperature and strain-rate dependence, and is put on a physical foundation in the framework of dislocation storage and dynamic recovery rates. The formalism can be applied to the steady-state limit of creep in the same range of temperatures and strain rates; the stress exponent of the creep rate must, as a consequence, be strongly temperature dependent, the activation energy weakly stress dependent. Near half the melting temperature, where available work-hardening data and available creep data overlap, they match. Extrapolation of the proposed law to higher temperatures suggests that no new mechanisms may be necessary to describe high-temperature creep. A new differential equation for transient creep also follows from the empirical work-hardening law.

Introduction

The work-hardening behavior of many materials as a function of temperature and strain rate is typified by experimental data displayed in Fig. 1, which were obtained for high-purity aluminum polycrystals. The qualitative characteristics are:

- 1 The annealed material shows a finite yield stress.
- 2 The initial work-hardening rate is insensitive to temperature and strain rate. (The "initial rate" is not operationally well defined at the higher temperatures, but the high-temperature results are consistent with the conclusions drawn from the lower-temperature observations.)
- 3 At larger strains, the work-hardening behavior depends strongly on temperature and strain rate.
- 4 At large strains, all curves tend to approach a finite "saturation stress." (This condition is not self-evident from the figure, especially for the lower temperatures; it will be more firmly established in the forthcoming.)

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The classical work-hardening law for engineering materials,

$$\sigma = K\epsilon^k \quad (1)$$

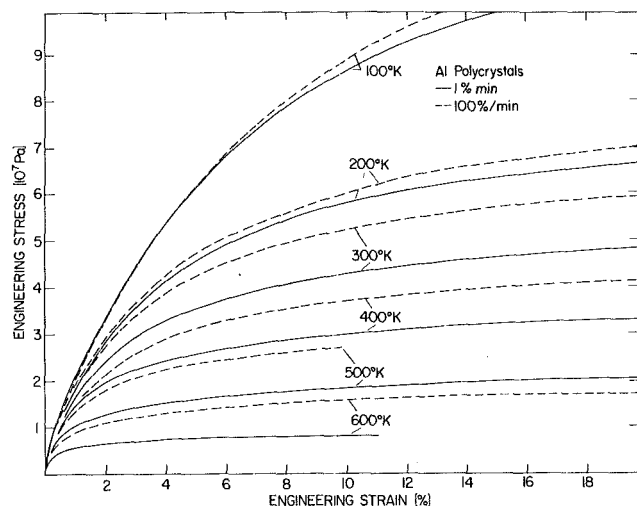


Fig. 1 Stress-strain curves for polycrystalline 99.99 percent pure aluminum at various temperatures and two strain rates [1]

fulfills none of these conditions. Whereas the first condition could be met by adding a further constant to the right-hand side of equation (1), conditions 2 and 3 cannot be met simultaneously, and condition 4 is grossly violated. It has become common to use different values of the two parameters of equation (1) for different ranges of the stress-strain curve: by letting either one get smaller one can begin to account for the tendency to saturation; and by letting them depend in different ways on temperature and strain rate, one may meet both conditions 2 and 3. At this point, the number of parameters would have become too large to provide a meaningful empirical description.

The tendency to flow-stress saturation (condition 4) is not a generally accepted fact, or is at least not usually emphasized. If it exists in true stress versus true strain diagrams, it cannot be absolutely established in tensile tests, because necking must intervene before saturation is reached. Compression tests have not been carried out to high enough strains, to the author's knowledge, but torsion tests have, in some cases, shown saturation. In the following section, we shall analyze tensile data and find that, the lower the work-hardening rate gets, the more plausible becomes the extrapolation to a zero value. We thus regard the existence of a saturation stress to be a basic characteristic of work-hardening behavior and call it σ_s .

The analysis further suggests that the simplest form of an approach to saturation, the exponential one, describes the data adequately well. In the upper range of the stress-strain curve, it can thus be described by the relation

$$\frac{\sigma_s - \sigma}{\sigma_s - \sigma_1} = \exp - \frac{\epsilon - \epsilon_1}{\epsilon_c} \quad (\sigma \geq \sigma_s/2). \quad (2)$$

This equation was first proposed in 1948 by Voce [2, 3]² who had also postulated the existence of a saturation stress. In addition, Voce pointed out that our foregoing condition 1 is fulfilled in an expanded and more basic sense by equation (2) than by equation (1): in two successive tests on the same specimen (by different investigators, for example) the "yield stress" in the second test (i.e., the stress at what the second investigator calls zero strain) must equal the stress that was reached in the previous test, but the subsequent work-hardening behavior should not depend on the yield stress of the original annealed material, or on the value of the strain measured in the first test. (See also Reed-Hill, et al. [4].) In equation (2), σ_1 is the measured yield stress in the present test (whatever the history of the specimen may be) and ϵ_1 is the arbitrary value of the strain at the beginning of the present test (commonly set equal to zero). This independence of the previous history makes it possible for us to consider the upper range of the stress-strain curve only, where we find equation (2) to be satisfied. To get the entire stress-strain curve, one must know more about the initial stage, which is in fact better understood from a mechanistic point of view, but is not of concern here.

In addition to the two initial values σ_1 and ϵ_1 , equation (2) contains two parameters: the saturation stress σ_s and a characteristic strain ϵ_c . Fig. 2 defines both parameters in a schematic diagram of the entire relation (2), with a hypothetical (but typical) "real curve" at lower strains. The figure also defines another parameter, which we shall use in preference to ϵ_c , namely the initial work-hardening rate Θ_0 ("initial" if equation (2) in fact held to zero stress). It is defined by the relation

$$\epsilon_c(T, \dot{\epsilon}) = \sigma_s(T, \dot{\epsilon})/\Theta_0. \quad (3)$$

Equation (3) specifies that while ϵ_c and σ_s depend on temperature and strain rate, the initial work-hardening rate Θ_0 will be found to be insensitive to these parameters and has a value of approximately $\mu/20$ to $\mu/15$ (where μ is the shear modulus).

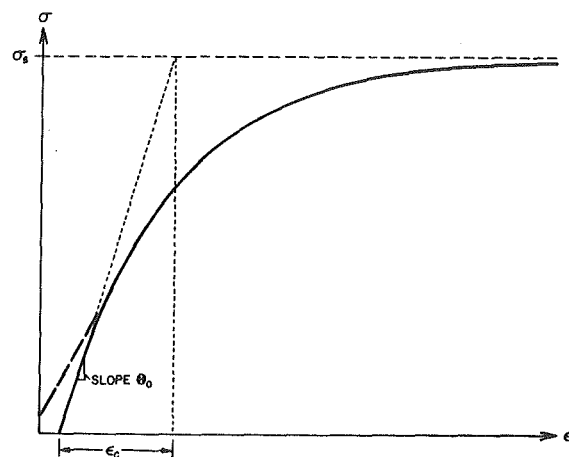


Fig. 2 Exponential saturation law for upper range of true stress-strain curve. For the lower range, a typical real curve is shown dashed.

Conditions 2 and 3 can thus be easily fulfilled simultaneously.

Stress-strain curves of the kind discussed here are obtained at constant strain rate. If a constant stress were in fact reached at a finite strain, the deformation would then precisely correspond to "steady-state creep": the constant-strain-rate response of the specimen under constant stress. The fact that, in most cases, the saturation stress in a test at constant strain rate is reached only asymptotically means that steady-state creep is also reached only asymptotically. Conversely, the observation that something very close to steady-state creep is in fact exhibited frequently, even under constant true stress, reaffirms the existence of a saturation stress in dynamic tests. The correspondence between the two test procedures that test the same specimen behavior can be used to predict steady-state creep laws from observed work-hardening behavior. This will be done in the last section and cast into the form conventionally used in creep research. There do not seem to be any glaring inconsistencies between these predictions and general observations, but creep results at less than half the melting temperature and the relatively high stresses discussed here are rare. Similarly, the approach to saturation in a dynamic test is equivalent to "transient" creep in the approach to steady state. Thus, our analysis of tensile test data has permitted us to predict a new transient creep law.

It must be emphasized that the tests discussed here are of the continuous kind: the "dynamic" tests are run at constant strain rate and the stress varies smoothly; the "creep" tests are run at constant (true) stress and the strain rate varies smoothly. The macroscopic behavior represents the combination of two effects: the change of dislocation structure with strain, and the dependence of the flow stress at a given structure on strain rate and temperature. These two effects may be separated by changing test conditions abruptly, such as making a strain-rate change in a dynamic test, or a stress change in a creep test. Such discontinuous tests are not described in this paper and are in fact not well understood in the high-strain range.

A mechanistic interpretation of the results will be presented in a central section. It is based on statistical dislocation storage and dynamic recovery. It will also provide new insight into the meaning of "dynamic recovery": the observed strain rate dependence of continuous work hardening precludes that it is simultaneous recovery of the conventional kind, namely a time-rate of change of structure; instead, "dynamic recovery" must be triggered by the passage of dislocations at recovery sites, and it is thus truly dynamic. All reasonable extrapolation to zero temperature suggests that it can be activated by stress alone.

²Numbers in brackets designate References at end of paper.

(The saturation stress extrapolated to 0 K is finite and of order $10^{-3}\mu$.) The temperature dependence of the saturation stress is compatible with a cross-slip model (but may well have more general validity).

An Empirical Work-Hardening Law

An analysis of work-hardening data is best performed on a differential basis, i.e., by plotting the work-hardening coefficient

$$\Theta = \left. \frac{\partial \sigma}{\partial \epsilon} \right|_{\epsilon, T} \quad (4)$$

against the true stress σ . This is less prejudicial towards a particular form of integrated stress-strain law, especially when there are different stages of work-hardening. It is also more fundamental, since the integrated strain (from the entire previous deformation history), which is operationally ill defined, is thus avoided. The current flow stress σ and the current work-hardening rate Θ , by contrast, are easily measurable in any arbitrary state of the material. The integrated stress-strain curve at constant strain rate and temperature can of course be obtained by integration from equation (4):

$$\epsilon = \epsilon_1 + \int_{\sigma_1}^{\sigma} \frac{d\sigma}{\Theta} \quad (5)$$

It is easily verified that the exponential saturation law proposed by Voce (equation (2)) has the differential form

$$\Theta = \Theta_0 \cdot \left(1 - \frac{\sigma}{\sigma_s} \right) \quad (6)$$

Fig. 3 shows the data of Fig. 1 converted to true stress and true strain and plotted in differential form. (The slopes were determined directly from the original curves shown in Fig. 1 and then multiplied by $(1 + e)^2$, whereas the ordinates in Fig. 1 were multiplied by $(1 + e)$ where e is the abscissa of Fig. 1.) An obvious feature of Fig. 3 is that the work-hardening rate decreases sharply with stress in all curves—the more sharply the higher the temperature and the lower the strain rate. In an intermediate range, this decrease may be approximately described by the linear form of equation (6). The range of good fit is substantial for the lower temperatures and becomes progressively smaller for the higher ones.

Fig. 3 shows significant deviations from the law (6) as the work-hardening rate gets very low. The data from tensile tests are known to become meaningless when the true work-hardening rate equals the true stress and necking instabilities develop. We believe that a change in mechanism must in fact occur in the case of aluminum (and in some others we have seen) already as the condition of constant applied force is approached. It seems to us inconceivable that otherwise the condition of constant force would be maintained for such large strains as is evident from Fig. 1. (The constant level in fact persists considerably beyond the strain values shown in the figure.) This behavior may well be related to one of the various phenomena described as "superplasticity." For this reason, we find it plausible to regard the lowest points on each curve as insignificant for the primary work-hardening behavior to be described here. Whether this is justified or not should be clearly established in compression tests carried to the same strain range.

Fig. 4 shows the same plot for data on type 304 stainless steel obtained by Diercks and Burke [5].³ They were evaluated beyond the necking condition, and it is seen that there is a clear break

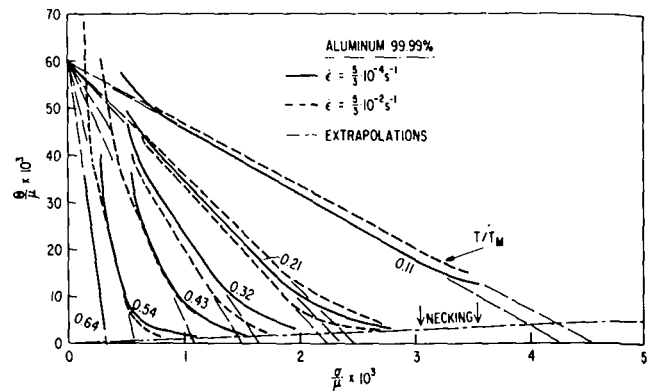


Fig. 3 Work-hardening coefficient Θ versus flow stress σ , both normalized by the temperature dependent shear modulus $\mu = \sqrt{C_{11}(C_{11}-C_{12})/2}$, for polycrystalline aluminum at various temperatures and two strain rates. Data from Fig. 1.

in the curves around the line $\Theta = \sigma$. Slight deviations from straight-line behavior are seen even before the necking condition is reached; they go smoothly into the behavior exhibited during necking. This reaffirms our belief that such deviations are due to a separate mechanism characteristic of tensile tests rather than of material behavior.

Fitting the midrange of the curves in Figs. 3 and 4 by straight lines, one may determine the parameters Θ_0 and σ_s in equation (6). The low-temperature data for aluminum, which are fitted well, extrapolate to an ordinate intercept Θ_0/μ that is insensitive to temperature and strain rate; this was so only when the work-hardening rate had been normalized by the temperature dependent shear modulus μ . (The particular combination of single crystal elastic constants used is suggested by dislocation theory; the polycrystal shear modulus could have been used almost as well; however, use of Young's modulus would bring in the temperature dependence of the bulk modulus which is irrelevant here.) Having established a tendency for the value of Θ_0/μ to be insensitive to temperature and strain rate, the other $\Theta(\sigma)$ curves for higher temperatures and lower strain rates were then made to extrapolate to the same value.

The stainless steel data in Fig. 4 do not extrapolate to a constant value of Θ_0/μ , although the intercepts scatter around the same value 0.06 as in aluminum (and copper). It is possible that in a commercial alloy like stainless steel one should account for the contributions to the flow stress from solution and precipitation hardening, for example by subtracting the yield stress (or a

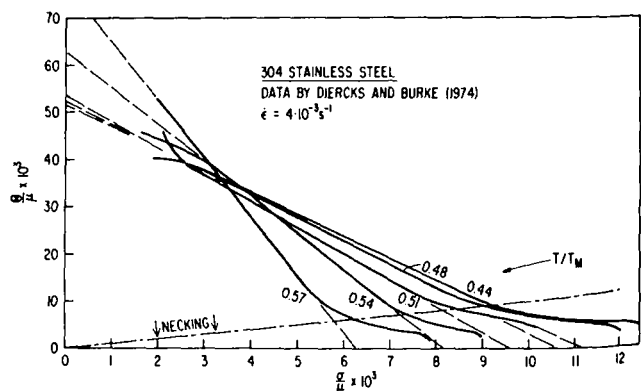


Fig. 4 Work-hardening coefficient Θ versus flow stress σ , both normalized by the temperature dependent polycrystal shear modulus μ , for type 304 stainless steel at five temperatures, after Diercks and Burke [5].

³Note added in proof: Very extensive data on type 304 stainless steel have been analyzed in a similar way and with essentially similar results by Swindeman [29].

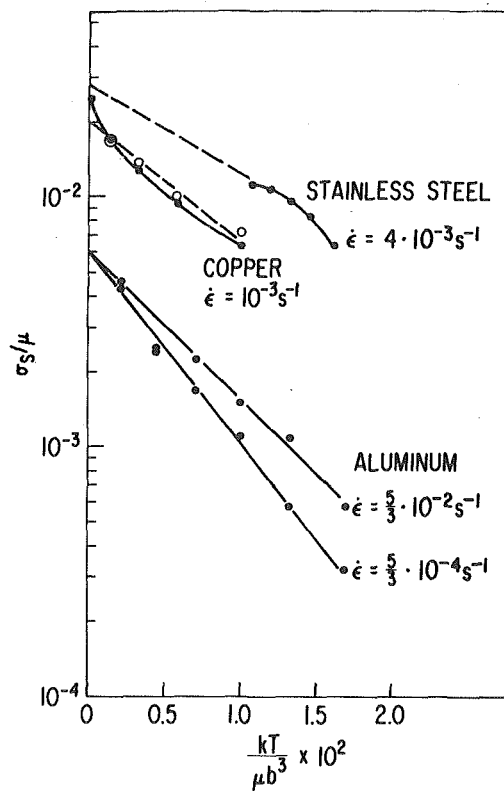


Fig. 5 Saturation stress σ_s , determined from Figs. 3, 4, and 8 for aluminum, stainless steel, and copper, plotted semilogarithmically versus temperature T . Both axes are normalized by the temperature dependent shear modulus μ (k = Boltzman constant, b = Burgers vector). The borders of the graph correspond approximately to the yield strength (for the pure material), the melting temperature, and the predicted maximum saturation stress, respectively.

portion of it) from all stresses. Fig. 4 suggests that such a procedure may in fact lead to a temperature independent value Θ_1/μ at this stress σ_1 . Equation (6) should then be amended to read

$$\Theta = \Theta_1 \cdot \left\{ 1 - \frac{\sigma - \sigma_1}{\sigma_s - \sigma_1} \right\} \quad (7)$$

(which is, of course, consistent with exponential saturation for any pair of initial values Θ_1 and σ_1). The stainless steel data used here are not sufficient to establish such a procedure for alloys as distinct from pure materials. For our evaluations, we shall continue to use equation (6), and essentially ignore the data for the highest temperature reported in Fig. 4.

The principal effect of temperature on the curves in Figs. 3 and 4 may be described by a strong temperature dependence of σ_s (and even of σ_s/μ). It is illustrated in Fig. 5, which also includes data for copper to be reported in the forthcoming. The combination of variables in the abscissa is chosen so as to

Table 1 Numerical data used for and obtained from evaluation of aluminum experiments reported in Figs. 1 and 3. $T_M=933\text{K}$: melting temperature; $b=0.286\text{nm}$: Burgers vector; $\mu = \sqrt{c_{11} \cdot (c_{11}-c_{12})/2}$: shear modulus, from Sutton [7].

$T[\text{K}]$	T/T_M	$\mu[\text{GPa}]$	$10^2 kT/\mu b^3$	$10^3 \sigma_s/\mu$ for $\dot{\epsilon}[\text{s}^{-1}] =$	
				$\frac{5}{3} \cdot 10^{-4}$	$\frac{5}{3} \cdot 10^{-2}$
100	0.107	28.1	0.209	4.26	4.54
200	0.214	26.7	0.441	2.36	2.48
300	0.321	25.3	0.697	1.65	2.24
400	0.429	23.8	0.991	1.09	1.50
500	0.536	22.4	1.316	0.57	1.09
600	0.643	20.9	1.691	0.32	0.57

Table 2 Numerical data used for and obtained from evaluation of stainless steel experiments reported in Fig. 4. $T_M=1700\text{K}$: assumed melting temperature; $b=0.252\text{nm}$: Burgers vector; μ : polycrystal shear modulus, as given by Diercks and Burke [5].

$T[\text{K}]$	T/T_M	$\mu[\text{GPa}]$	$10^2 kT/\mu b^3$	$10^3 \sigma_s/\mu$
755	0.44	60.7	1.07	11.1
811	0.48	58.7	1.19	10.7
866	0.51	56.5	1.32	9.6
922	0.54	54.6	1.45	8.2
977	0.57	52.4	1.61	6.3

Table 3 Numerical data used for and obtained from evaluation of copper data reported in Figs. 7 and 8. $T_M=1356\text{K}$: melting temperature; $b=0.255\text{nm}$: Burgers vector; $\mu = \sqrt{c_{11} \cdot (c_{11}-c_{12})/2}$, from Overton and Gaffney [8] and Chang and Himmel [9].

$T[\text{K}]$	T/T_M	$\mu[\text{GPa}]$	$10^2 kT/\mu b^3$	$10^3 \sigma_s/\mu$	
				at T	at 77K
4.2	0.003	45.8	0.0076	25.0	
77	0.057	45.4	0.140	17.0	17.0
173	0.128	44.0	0.326	12.8	13.9
293	0.216	42.1	0.574	9.35	9.90
473	0.349	39.2	0.997	6.35	7.20

Table 4 Summary of work-hardening parameters derived from Figs. 3 through 8

	Aluminum 99.99%	Copper 99.999%	Stainless steel type 304
$\Theta_0/\mu \times 10^2$	6.0	6.0	~ 5.5
$\sigma_{s0}/\mu \times 10^2$	0.6	2.0	~ 2.8
$\dot{\epsilon}_0[\text{s}^{-1}]$	$5 \cdot 10^4$	$5 \cdot 10^{4(b)}$	$5 \cdot 10^{4(b)}$
$\mu b^3/A$	9.0	$6.3^{(a)}$	~ 5.0

(a) From data on flow stress at $T = 77\text{K}$ after deformation at higher temperatures: using A' of equations (29) and (30) rather than A

(b) Assumed values

eliminate another trivial influence of the temperature dependence of μ , which is well known from thermal-activation theory (see, e.g., Kocks, et al. [6]). In the semilogarithmic plot of Fig. 5, the aluminum data are well described by straight lines of two different slopes for the two different strain rates. The data for copper and stainless steel, which extend over a smaller temperature range, are consistent with such an interpretation, although they might not by themselves have led to it. The deviations from straight-line behavior are in fact to be expected in the directions shown, as will be discussed later. All data used and derived are listed in Tables 1 through 4.

Straight lines in Fig. 5 may be represented by the following equation:

$$\ln \frac{\sigma_s}{\sigma_{s0}} = - \frac{kT}{A} \ln (\dot{\epsilon}_0/\dot{\epsilon}) \quad (8)$$

where k is the Boltzmann constant and two of the empirical parameters, σ_{s0} and A , contain an implicit temperature dependence proportional to $\mu(T)$. The form of the strain-rate dependence shown in equation (8) is, of course, not warranted by the data presented here; it is an educated guess. The form of the relation between saturation stress and temperature, i.e., the choice of a semi-logarithmic plot for Fig. 5, was suggested by an analogy to the temperature dependence of the stress at the beginning of stage III work hardening in single crystals [10], which bears considerable resemblance to the polycrystal behavior described here.

When data for both different temperatures and different strain rates are available, all constants in equation (8) can be deter-

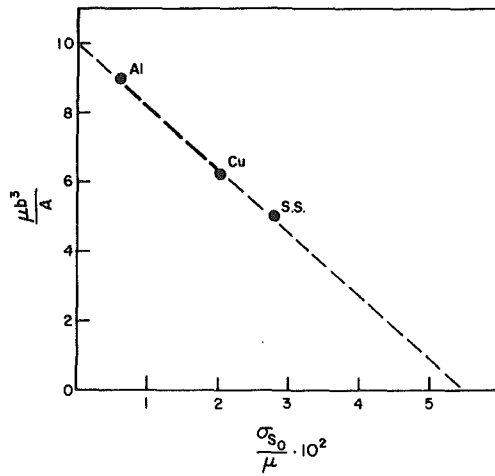


Fig. 6 Correlation between the slope and the intercept in Fig. 5 for different materials; it is interpreted in terms of the stacking-fault energy

mined. For aluminum, a comparison of the two slopes in Fig. 5 that relate to two strain rates a factor 100 apart, gives a value $A = \mu b^3/9$. The absolute value of either slope then gives $\dot{\epsilon}_0 \approx 5 \cdot 10^4 \text{ s}^{-1}$. For copper and stainless steel, we have measurements for only one strain rate. Assuming arbitrarily that the value of $\dot{\epsilon}_0$, to which the results are not very sensitive, is the same for these materials, we can derive values of A from the absolute values of the slopes for copper and stainless steel. Table 4 summarizes the numbers derived from the evaluation.

We now combine equations (6) and (8) into a single differential work-hardening law for the upper range of the stress-strain curve:

$$\Theta = \Theta_0 \cdot \left\{ 1 - \frac{\sigma}{\sigma_{s0}} \cdot \left(\frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right)^{kT/A} \right\} \quad \Theta \leq \Theta_0/2 \quad (9)$$

In addition to the explicit temperature dependence, there is an implicit one through the shear modulus, which enters the parameters Θ_0 , σ_{s0} , and A ; otherwise, these parameters and $\dot{\epsilon}_0$ are constants over the applicable range.

Equation (9) contains four parameters: one for each variable. This is convenient on dimensional grounds, but is one more than can be determined from experiments. In fact, Fig. 5 suggests that there may be a relation between σ_{s0} and A : the curves with the higher intercepts also have the lower slopes. The sequence for both parameters is the same as that of the moduli (which should have been normalized out already), of the melting points and the self-diffusion activation energies (which should be irrelevant for σ_{s0} at zero temperature), and of the stacking-fault energies. The latter would be expected, in the direction observed, according to the cross-slip theory of stage III deformation, which led us to expect an equation of the form (8) in the first place. It predicts linear relations in terms of the stacking-fault energy χ , of the form [11, 12]:

$$\sigma_{s0}/\mu = C_1 - C_2\chi/\mu b \quad (10)$$

$$\frac{\mu b^3}{A} = C_3 + C_4\chi/\mu b \quad (11)$$

Thus, a correlation between the experimental values of σ_{s0} and A should be attempted in a plot like Fig. 6. The two points for aluminum and copper were used to obtain a straight line according to equations (10) and (11). The values for stainless steel can only be obtained by relatively far extrapolations in both Fig. 5 and Fig. 6; they were *picked* such as to make *both* look reasonable. It may be mentioned in passing that we have ob-

tained similar plots for single crystals of aluminum, copper, silver, and gold; they all fall within this scheme (Mecking and Kocks, unpublished).

If the linear relation of Fig. 6 held through the entire range, it would predict an upper limit for $\sigma_{s0} = C_1$ for zero stacking-fault energy. An upper limit for C_1 is obtained by observing that $C_4 \geq 0$; this was used to define the top edge of Fig. 5. It is about half the ideal shear strength.

In summary, the empirical work-hardening law (9) has the following general features:

- 1 The work-hardening rate tends to a finite value Θ_0 at zero stress, which is between $\mu/20$ and $\mu/15$ for the materials that have been analyzed. (In most materials, particularly in single crystals, the maximum work-hardening rate observed is in fact somewhat smaller and constant over a finite temperature range; this corresponds to the qualitative behavior depicted in Fig. 2.)
- 2 There is a saturation stress, at which the work-hardening rate tends to zero, even at zero temperature. Its value σ_{s0} is independent of strain rate and varies between about 0.6 and $3 \times 10^{-2} \mu$ for different materials, the higher values being those for materials with lower stacking-fault energy.
- 3 The saturation stress is lowered by increasing the temperature and, at any finite temperature, by decreasing the strain rate. The effect of temperature is the more pronounced the higher the stacking-fault energy.

A Mechanistic Interpretation

It is well understood that plastic deformation in materials of cubic lattice structures (whether they be single crystals or polycrystalline aggregates) occurs by dislocation glide, at least in the temperature range from about 0.1 to 0.5 T_M . At lower temperatures, deformation twinning may sometimes occur; at higher temperatures, diffusion-controlled mass flow and some grain-boundary sliding may occur in polycrystals. Dislocation climb may well occur over part of the temperature range of interest here; it does not significantly contribute to the deformation, but may be an important mechanism in the recovery processes to be dealt with in the forthcoming.

In all dislocation models, work hardening is a consequence of the fact that some fraction of the mobile dislocations that produce the strain do not exit through the surface, are not absorbed in grain boundaries, and do not annihilate each other, but are stored in the crystals or react with other dislocations to form new obstacles to continued slip. If we call τ the component of the flow stress due to obstacle dislocations (which may be superimposed, in one way or another, on other flow-stress contributions), it is linked to the density per unit area, ρ , of obstacle dislocations by the relation

$$\tau = \alpha \mu b \sqrt{\rho}. \quad (12)$$

Here, b is the amount of the Burgers vector and α a proportionality constant of order 1. The shear modulus μ is a particular combination of single-crystal elastic constants, which may usually be adequately described by the polycrystal shear modulus.

The change in dislocation density may be considered to consist of two components:

$$d\rho = d\rho_{\text{STOR}} - d\rho_{\text{RECOV}}. \quad (13)$$

The first term describes a change in obstacle density due to the fact that a certain fraction of the mobile dislocation density ρ_m has been stored in the crystal after it has moved through a distance dx . Defining a mean free path Λ , we can write this term as

$$d\rho_{\text{STOR}} = \rho_m \cdot \frac{dx}{\Lambda} \quad (14)$$

If the mean free path is determined by a statistical process as the dislocation moves through the obstacle dislocations (and no inclusions are present, for example), it is proportional to the

average spacing between obstacle dislocations:

$$\Lambda = \beta/\sqrt{\rho}. \quad (16)$$

A second contribution to the change in dislocation density may be due to some recovery or rearrangement process that occurs among the dislocations already stored previously. These recovery processes are not well understood. Various different mechanisms have been proposed for them. In the following, we will outline a mechanism that does lead to the observed work-hardening law (9) and can be rationalized a posteriori.

The number of potential recovery sites in an area element da of the slip plane is ρda . When an average length L_R of dislocation gets annihilated or becomes ineffective in some other way at each potential recovery site, the total change in dislocation length in the volume V is

$$V \cdot d\rho_{\text{RECOV}} = L_R \cdot \rho da. \quad (17)$$

The essential assumption that must now be made to give the desired answer is that the probability of a recovery event is proportional to the number of times that a potential recovery site is contacted by a moving dislocation. Then, the slip-plane area da , as well as the slip distance dx previously used, are linked to the shear strain increment

$$d\gamma = \rho_m b dx = b da/V. \quad (18)$$

Combining equations (14) through (18) we find

$$\frac{d\rho}{d\gamma} = \frac{\sqrt{\rho}}{\beta b} - \frac{L_R}{b} \cdot \rho. \quad (19)$$

Finally, if we express the dislocation density ρ by the flow stress τ , from equation (12), we obtain

$$\theta \equiv \frac{d\tau}{d\gamma} = \frac{\alpha\mu}{2\beta} - \frac{L_R}{2b} \cdot \tau. \quad (20)$$

This equation differs fundamentally from all previous recovery theories which have been expressed in the form (see, e.g., Kocks [13])

$$\theta = h - \frac{r(\tau, T)}{\dot{\gamma}}. \quad (21)$$

The explicit dependence on $1/\dot{\gamma}$ in this equation comes from the conventional assumption that the change in flow stress due to recovery is proportional to time. Since the observed dependence of the work-hardening rate on the strain rate is considerably less than that expressed by equation (21), we have found it necessary to adopt a "mechanical" recovery model (or a truly dynamic one): although the recovery rate may be influenced by thermal activation, it is a priori a rate of change with strain, not with time. That this rate of change of structure is proportional to the current level of the structure itself, which seems an obvious first-order assumption, has led to the linear dependence on the flow stress in equation (20).

Equation (20) may be condensed into the form

$$\theta = \theta_0 \cdot \left(1 - \frac{\tau}{\tau_0}\right) \quad (22)$$

which bears an obvious similarity to equation (6). There are, however, nontrivial differences between the τ and γ used here and the σ and ϵ used previously. First, there is the relatively trivial difference that τ and γ here refer to resolved shear stress and shear, respectively, in a single set of crystallographic slip planes, whereas σ and ϵ refer to the stress and strain in the macroscopic coordinate system on a polycrystalline material. Let us describe the ratios σ/τ and $d\gamma/d\epsilon$ by an effective orientation factor M ; for materials of the face-centered cubic lattice structure in tension or compression, $M = 3.06$ according to Taylor and later investigators (see, e.g., Kocks [14]). In addition,

we must observe the following.

The shear increment $d\gamma$ used here is strictly that due to long-range dislocation glide. An operationally defined macroscopic strain increment $d\epsilon$ contains in addition elastic as well as anelastic and plastic strain increments that are proportional to the stress increment and are important during the loading transition only. (For a more extensive treatment of these effects, see Kocks [15]). Let us lump all these effects together into an effective Young's modulus E_{eff} ; then

$$\dot{\epsilon} = \dot{\gamma}/M + \dot{\sigma}/E_{\text{eff}}. \quad (23)$$

Finally, the flow-stress τ defined in equation (12) is strictly a function of the structure of the material (and other material properties), whereas the stress used in the previous section was a measurable force per unit area on the specimen. These two become synonymous if one observes two important rules: (1) one measures the particular value of the applied stress at which the loading transients have been overcome and true plastic behavior has started—e.g., by back-extrapolating the plastic stress-strain curve on a suitable plot to the intersection with the elastic line; (2) one eliminates effects of temperature and strain rate either by extrapolating the flow stress to zero temperature or—more practically—by confining oneself to a single standard of temperature and strain rate. Inasmuch as effects of different strain rates and temperatures are of prime concern in our present discussion, we must incorporate them into the definition of flow stresses, and shall do so in the following way. We continue to call τ the flow stress at any selected standard of temperature and strain rate, but allow σ to be the flow stress (in the sense defined in the foregoing condition 1) at any arbitrary macroscopic strain rate and temperature.

Then,

$$\sigma(\dot{\epsilon}, T) = M \cdot \tau \cdot s \left(kT \ln \frac{\dot{\epsilon}_0}{\dot{\epsilon}} \right). \quad (24)$$

where the argument of the function s is the combination expected from thermal activation theory. The function s describes the ratio of flow stresses at different temperatures and strain rates at the same structure (the Cottrell-Stokes ratio). It is common to define the "isostructural" strain-rate sensitivity by the phenomenological parameter

$$m \equiv \frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \bigg|_{T, \text{ structure}} \quad (25)$$

(The inverse of this quantity is called m in part of the literature.) The relation between this parameter m and the derivative of the function s is:

$$\frac{s'}{s} = - \frac{1}{mkT} \quad (26)$$

With these definitions, the macroscopic work-hardening rate Θ used in the preceding section and the microscopic coefficient θ in this one becomes

$$\Theta(\sigma, \dot{\epsilon}, T) = M^2 \cdot s \cdot \theta \left(\frac{\sigma}{M\tau}, M\dot{\epsilon}, T \right). \quad (27)$$

Note that even the quantity θ on the right-hand side may, in general, depend on strain rate and temperature. This may be illustrated by the following precise description of how this quantity is measured. Pick two sets of the pair of variables $\dot{\gamma}$ and T : the "standard" conditions, at which the structure parameter τ (equation (12)) is measured, and the "test conditions," at which the material is being strained. Deform the material to a strain γ_1 and a stress σ_1 under the test conditions, then take it to the standard conditions and find a flow stress $M\tau_1$. Take the material back to the test conditions and deform it by an incremental amount to a new strain γ_2 , observing a new stress

σ_2 . Returning to the standard conditions, find a new flow stress $M\tau_2$. The work-hardening rate $\theta = (\tau_2 - \tau_1)/(\gamma_2 - \gamma_1)$, while $\Theta = (\sigma_2 - \sigma_1)/(\epsilon_2 - \epsilon_1)$.

There is no a priori reason why θ should not depend on the test conditions (in addition to the standard conditions). In fact, it did in the experiments on copper polycrystals by Bullen and Hutchison [16]. They used precisely the procedure described in the foregoing, using standard conditions of $T = 77\text{K}$ and $\dot{\epsilon} = 10^{-3}\text{ s}^{-1}$. The data from some of their test temperatures have been replotted in Fig. 7. The continuous curves are interpolated from the $\sigma(T)$ data in Fig. 1 of Bullen and Hutchison [17], the circles represent the flow stresses observed under standard conditions according to Fig. 2 of Bullen and Hutchison [16]. Whereas Bullen and Rogers [18] interpreted these data in terms of the power law (equation (1)), Fig. 8 displays our evaluation of the derivatives. (See also Bocek [19].) It is seen that $\theta(\tau)$ has the same form as $\Theta(\sigma)$. Pairs of points at the same strain should be linked by lines that extrapolate to the origin, as specified by equation (27). The scatter is much too large to allow such a conclusion, but the data are consistent with it.

Extrapolating the lines $\theta(\tau)$ to $\theta = 0$, one finds $M\tau_s$, and may plot it against temperature, along with σ_s , as it was done in Fig. 5 (circles). The linear law (8) is obeyed better for τ_s than it is for σ_s . This is not surprising. Parameter comparison between equations (6) and (22), using (24) and (27), shows that

$$\sigma_s = M \cdot \tau_s \cdot s. \quad (28)$$

The saturation stress σ_s has an additional dependence on temperature (and strain rate) to that of τ_s , namely, the Cottrell-Stokes ratio. It is known to decrease slightly with increasing temperature at low temperature (explaining the copper data in Fig. 5), be constant over a large range of intermediate temperatures (explaining the aluminum data), and decrease rapidly at higher temperatures (explaining the curvature of the stainless-steel data in Fig. 5).

We are thus led to hypothesize that the temperature and strain-rate dependence of the saturation stress that was specified in equation (8) holds even more accurately and over a wider temperature range for τ_s . This may be written as

$$\frac{\dot{\gamma}}{\dot{\gamma}_0} = \exp - \frac{A' \cdot \ln(\tau_{s0}/\tau_s)}{kT}. \quad (29)$$

It may be possible to rationalize this equation in the following way. The right-hand side is a well-known approximate description of the cross-slip probability at the stress τ_s [20]. It is in fact quite possible that it is an equally adequate approximate description of other processes that might help recovery, such as climb. The equality (29) could then mean that, at τ_s , this probability equals the rate of arrival of dislocations at recovery sites, which is proportional to the left-hand side. Extension of this argument to stresses other than τ_s would lead to the conventional theories of a very strongly rate sensitive recovery rate that is, specifically, at the basis of the dynamic recovery theory of Lücke and Mecking [21]. In our heuristic theory, it is meant to apply only at τ_s ; the work-hardening rate at all other stresses is described by

$$\theta = \theta_0 \cdot \left\{ 1 - \frac{\tau}{\tau_{s0}} \left(\frac{\dot{\gamma}_0}{\dot{\gamma}} \right)^{kT/A'} \right\} \quad (30)$$

which follows from equations (22) and (29).

Implications for Low-Temperature Creep

The tests discussed until now were of the "dynamic" kind, where a constant strain rate is prescribed and the stress response is measured. In "creep" experiments on the other hand, the (true) stress is applied and the strain-rate response is measured. This strain-rate response usually has a stage called "secondary

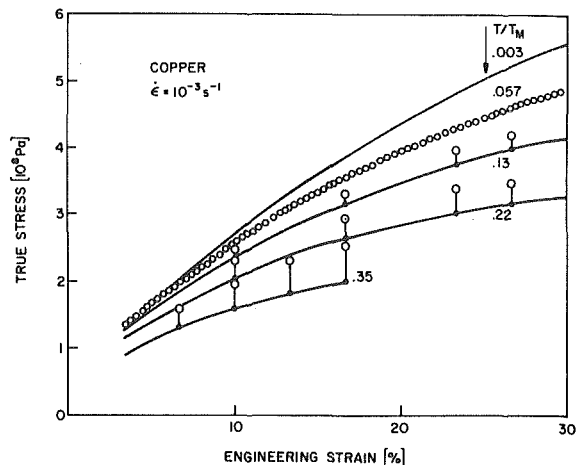


Fig. 7 Stress-strain curves for polycrystalline high-purity copper, after Bullen and Hutchison [17]; also (o) flow stresses at $T = 77\text{K}$ after various histories (Bullen and Hutchison [16])

creep" or "steady-state creep" where the strain-rate response is constant. In some cases, this constant strain rate merely corresponds to an inflection point in the strain-time diagram, in others it is more like asymptotic behavior. At low temperatures, a secondary stage is usually not measured; at high temperatures on the other hand, steady state is often observed to exist over large ranges of strain.

In the temperature range of interest here, we have found that dynamic tests at a prescribed strain rate $\dot{\epsilon}$ do tend to asymptotically approach a constant stress σ_s . In this limit, the question of what is prescribed and what is the response becomes irrelevant as far as the specimen behavior is concerned: if a stress σ equal to the saturation stress were in fact prescribed (by an applied weight, say), the strain-rate response would have to tend to the same strain rate that was previously prescribed in the dynamic test. This tendency should be an asymptotic one, if the work-hardening law (6) is obeyed, and true steady state should never be reached. The asymptotic steady-state strain rate $\dot{\epsilon}_s$ is, however well defined and must be (equation (8)):

$$\dot{\epsilon}_s = \dot{\epsilon}_0 \cdot \left(\frac{\sigma}{\sigma_{s0}} \right)^{A/kT} \quad (31)$$

This is the conventional power law for creep, with the "stress exponent"

$$n = \frac{\partial \ln \dot{\epsilon}_s}{\partial \ln \sigma} \bigg|_T = \frac{\partial \ln \dot{\epsilon}}{\partial \ln \sigma} \bigg|_T = \frac{A}{kT} \quad (32)$$

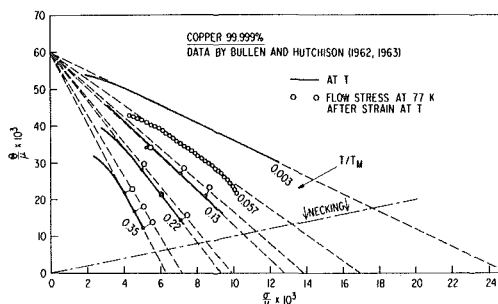


Fig. 8 Work-hardening coefficient Θ versus flow stress σ ($\circ$ at temperature, $\circ$ at 77K), both normalized by the temperature dependent shear modulus $\mu = \sqrt{c_{11}(c_{11}-c_{12})/2}$, for polycrystalline high-purity copper. Data from Fig. 7.

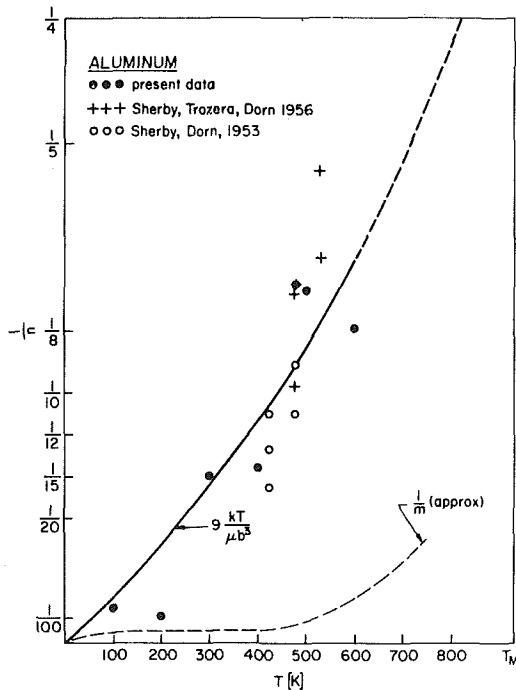


Fig. 9 The stress exponent n of the steady-state creep rate, plotted inversely versus temperature. The solid line and the points follow from our evaluation of work-hardening data, the crosses and circles from literature data on creep. The lower dashed line is an estimate of the strain-rate sensitivity of the instantaneous flow stress, which should be a component of $1/n$ in the solid line and should, therefore, change its extrapolation to higher temperatures significantly.

or, using equations (25) through (29),

$$\frac{1}{n} = \frac{kT}{A'} + \frac{1}{m} \quad (33)$$

The steady-state stress exponent n must therefore be strongly temperature dependent; in addition to the explicit temperature dependence shown in equations (32) and (33), A' (or A) depends on temperature through the modulus, and the isostructural stress exponent m changes with temperature both directly and through the applied stress at a given strain rate.

Fig. 9 shows $1/n$ versus T , the solid line representing equation (32) over the measured range. In this range, the contribution from $1/m$, which is shown qualitatively as a dashed line, is seen to be negligible; for an extrapolation to higher temperatures, however, it may be significant. The points in Fig. 9 represent the values of n derived from pairs of points in Fig. 5 at the two strain rates investigated for each temperature. The circles and crosses are values of n derived from creep experiments by Sherby and Dorn [22] and Sherby, Trozera, and Dorn [23], respectively. Within the considerable scatter (which is to be expected from such measurements), the agreement is certainly satisfactory. For a wider range of temperatures and materials, Blum and Reppich [25] have evaluated data from the literature and found a linear correlation between n and σ/T (most of the variation coming from variations in temperature rather than stress), which they interpret in the framework of a different model.

It is interesting to speculate on the behavior at temperatures higher than those reported here. An extrapolation of the solid line in Fig. 9, with or without adding the lower dashed line for $1/m$, gives values for n between 5 and 4 for temperatures between 650 and 800K, or $T/T_M = 0.7$ to 0.85. Such stress exponents are commonly quoted for aluminum, and the commonly used temperatures for steady-state creep measurements are

indeed in this range. It thus appears possible that no new mechanism is required to explain high-temperature creep, in addition to the one causing low-temperature behavior and indeed extending to $T = 0K$. Such speculation is reinforced by the observation that equations (32) and (11) predict a dependence of the stress exponent on the stacking-fault energy in the direction known to hold from many experiments (see, e.g., Bird, Mukherjee, and Dorn [25]). This has always been hard to justify on the basis of a diffusion controlled mechanism.

One may also define, in the usual way, an effective activation energy for steady-state creep

$$Q_s = \left. \frac{\partial \ln \dot{\epsilon}_s}{\partial (-1/kT)} \right|_{\sigma} \quad (34)$$

which is the activation enthalpy ΔH_s if $\dot{\epsilon}_0$ is independent of temperature. Under the same assumption and, additionally, the hypothesis that all entropy effects are due to the variation of the shear modulus with temperature, the activation free enthalpy for steady-state creep is

$$\Delta G_s = \mu b^3 \cdot \left. \frac{\partial \ln \dot{\epsilon}_s}{\partial (-\mu b^3/kT)} \right|_{\sigma/\mu} \quad (35)$$

(See, e.g., Kocks, et al., p. 135 [6]). This quantity follows directly from the present equation (8) or (31):

$$\Delta G_s = A \cdot \ln (\sigma_0/\sigma) \quad (36)$$

if the parameters are evaluated from the normalized plot in Fig. 5. It is seen that the activation energy must be stress dependent, although rather weakly. Conversely, one may express the activation energy in terms of temperature and strain rate:

$$\Delta G_s = kT \cdot \ln (\dot{\epsilon}_0/\dot{\epsilon}) \quad (37)$$

where again the value of $\dot{\epsilon}_0$ must be the one derived from a normalized plot like Fig. 5, here 5.10^4 s^{-1} . This relation is plotted in Fig. 10 for aluminum as a solid line.

What has usually been measured in creep experiments is not this quantity but Q_s as given by equation (34). It is always larger—the more so the higher the temperature. The one value plotted as a cross in Fig. 10 is such a measured value of Q_s and is thus not necessarily in disagreement with our prediction on ΔG_s ; it was given by Sherby, Trozera, and Dorn [23] from a single comparison of two creep curves at the same stress.

Let us again speculate on the behavior at higher temperatures and lower strain rates, shown dashed in Fig. 10. The common practice in this range is to compare the measured activation energy to that of self-diffusion. To provide a check on the rate-controlling mechanism, the proper quantities to be compared are the activation free enthalpies ΔG . Diffusion experiments yield the activation enthalpy ΔH_{SD} , but current diffusion theory sees little difference between that and ΔG_{SD} . The difference should be substantial in creep experiments (see, e.g. Kocks, et al., p. 240 [6]). With currently available data, no meaningful comparison can be made. Fig. 10 shows that the range of self-diffusion activation energies that have been measured for aluminum [26] and are indicated as a shaded area, overlaps our values over a significant temperature range.

This agreement may, of course, be fortuitous. Indeed, it would be surprising if diffusion played no role at all in high-temperature creep. Just as surprising it would be if diffusion were rate controlling at the low temperatures investigated here. Yet the data available until now do not force one, from a phenomenological point of view, to invoke more than one mechanism for the entire temperature range. It is a challenging theoretical task to find a mechanism that smoothly spans low and high temperatures.

Until now, we have discussed only the steady-state limit of creep. The approach to the steady-state asymptote in "transient creep" should be an expression of the same material charac-

teristics that determine the *approach* of stress-strain curves to saturation under constant strain rate. We should thus be in a position to use the derived empirical work-hardening law in order to establish an empirical law for transient creep.

What makes the strain rate vary in a constant stress test is that the structure changes. We have expressed the structure, in equations (12) and (24) by the parameter τ , the flow stress measured under some standard set of strain rate and temperature. For the purposes of the approach to steady-state creep, it is opportune to choose as the standard conditions the strain rate actually expected in the steady-state asymptote at that temperature and stress. The structure parameter τ could then, in principle, be determined by interrupting a transient creep test periodically by a short dynamic test at this higher strain rate. The ratio of the flow stress $M\tau$ to the applied stress σ should then be given by

$$\left(\frac{M\tau}{\sigma} \right) = \left(\frac{\dot{\gamma}_s}{\dot{\gamma}} \right)^{1/m} \quad (38)$$

and the rate of flow-stress change with strain, i.e., the work-hardening law can be written as

$$\theta = \theta_0 \cdot \left\{ 1 - \frac{M\tau}{\sigma} \right\} \quad (39)$$

or, with equation (25),

$$\theta = \theta_0 \cdot \left\{ 1 - \left(\frac{\dot{\gamma}_s}{\dot{\gamma}} \right)^{1/m} \right\}. \quad (40)$$

What is observed in transient creep is a progressive decrease of the strain rate with strain, which is best described (see, for example, Sherby, Lytton, and Dorn [27]), by

$$-\frac{\partial \ln \dot{\gamma}}{\partial \gamma} \bigg|_{\sigma, T} = m \frac{\theta}{\tau}. \quad (41)$$

Inserting equation (37) we find the newly proposed empirical differential equation for transient creep:

$$-\frac{\partial \ln \dot{\gamma}}{\partial \gamma} \bigg|_{\sigma, T} = m \frac{\theta_0}{\tau} \cdot \left\{ 1 - \left(\frac{\dot{\gamma}_s}{\dot{\gamma}} \right)^{1/m} \right\}. \quad (42)$$

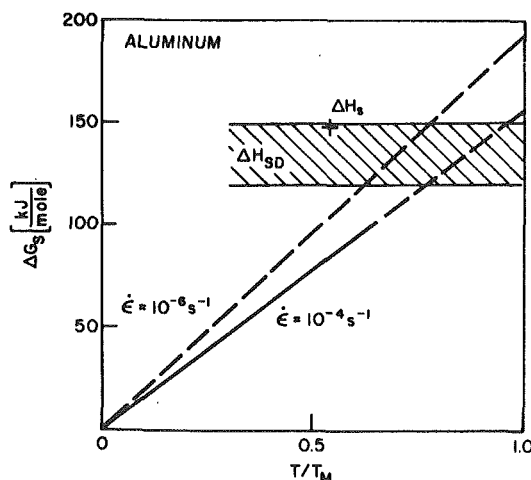


Fig. 10 Activation free enthalpy ΔG_s for steady-state creep as a function of temperature for aluminum. Solid line: prediction from work-hardening data; dashed lines: extrapolations from this prediction; +: a value of ΔH_s ($= 34$ kcal/mole) given by Sherby, Trozera, and Dorn [23]; shaded area: range of values for the self-diffusion activation enthalpy ΔH_{SD} (Bishop and Fletcher, [25]).

It obviously has the right limiting behavior, but has not been tested experimentally.

It must be emphasized that the correspondence between dynamic tests and creep tests can, of course, only be expected when the range of the variables is similar. In practice, they rarely are: creep tests are usually done at strain rates three or four orders of magnitude lower than dynamic tests. One can estimate whether this is important or not in the following way. The deformation maps published by Ashby [28] show, for temperatures below about half the melting point, three possible ranges of mechanisms. At the low-stress extreme is grain-boundary transport ("Coble creep"); it is never important at strain rates above about 10^{-8} s^{-1} and shall be excluded from consideration here. At the high-stress extreme is what Ashby calls "glide" and includes what we have discussed here. The intermediate range is labeled "low-temperature creep" by Ashby and identified with dislocation climb controlled by core diffusion. We find it plausible that this latter range is in fact indistinguishable from dynamic recovery in glide, whatever that may be due to. In both regions, the stress exponents n should be large compared to 1; this distinguishes them operationally from Coble creep. We should thus use the following rule: If the stress exponent n observed in actual creep experiments is, say, 5 or more, the mechanism is intimately connected with dislocation glide and will not qualitatively change by further increases in strain rate; it should thus be the same as that observed in dynamic tests. More data need clearly be taken and analyzed to establish this rule more firmly.

Summary

It is shown that the true work-hardening rate decreases approximately linearly with the flow stress over a significant range of the stress-strain curve, almost to the necking instability. It extrapolates to zero at a finite "saturation stress." The logarithm of this saturation stress decreases approximately linearly with temperature and logarithmically with strain rate. This corresponds to a power law for the relation between applied stress and steady-state creep rate in which the exponent depends linearly on temperature. An application of the work-hardening law in its approach to saturation implies a new transient creep law in its approach to steady state.

The parameters that characterize material behavior in this description are as follows: an initial work-hardening rate θ_0 (extrapolated to zero flow stress), which is insensitive to temperature and strain rate and has a value of about 1/20 to 1/15 of the shear modulus for the materials investigated; a mechanical saturation stress σ_0 (which would be approached at absolute zero temperature) of the order of 1/100 of the shear modulus, apparently increasing with decreasing stacking-fault energy; and two parameters A and $\dot{\epsilon}_0$, which normalize the temperature and strain-rate dependence, respectively; A is correlated with σ_0 and also rises as the stacking-fault energy falls.

Data on polycrystalline high-purity aluminum, high-purity copper, and commercial type 304 stainless steel, spanning the range of temperatures up to about $0.6 T_M$, have been evaluated and have shown the behavior described in the foregoing, which was first abstracted from data on single crystals of copper, silver, gold, and aluminum. Extrapolation of the proposed relations to higher temperatures is consistent with generally observed behavior without involving additional mechanisms.

An interpretation of these empirical laws in terms of dislocation mechanisms leads one to postulate a recovery rate that is proportional to the obstacle density, which seems reasonable, and to strain rather than time elapsed, which is unconventional. It suggests that dynamic recovery is activated by the passage of mobile dislocations past recovery sites, is possible at zero temperature, and is helped by a thermally activated mechanism such as cross slip.

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